

Experiment 15. Implement Probabilistic Context-Free Grammar (PCFG) Parsing

Program

```
grammar = {  
    ("John", "runs"): 0.95,  
    ("John", "walks"): 0.80,  
    ("Mary", "runs"): 0.90,  
    ("Mary", "walks"): 0.85  
}  
  
sentence = tuple(input("Enter sentence: ").split())  
  
if sentence in grammar:  
    print("Sentence Accepted")  
    print("Probability:", grammar[sentence])  
else:  
    print("Sentence Rejected")
```

Output

```
0] # Simple PCFG Parser
19s

grammar = {
    ("John", "runs"): 0.95,
    ("John", "walks"): 0.80,
    ("Mary", "runs"): 0.90,
    ("Mary", "walks"): 0.85
}

sentence = tuple(input("Enter sentence: ").split())

if sentence in grammar:
    print("Sentence Accepted")
    print("Probability:", grammar[sentence])
else:
    print("Sentence Rejected")

... Enter sentence: John runs
Sentence Accepted
Probability: 0.95
```

✓ 14:38 Python 3

Windows taskbar: Search, File Explorer, WhatsApp (35), Edge, VS Code, Teams, Chrome, Firefox, Word. System tray: ENG IN, 14:39, 28-07-2026.